Unsupervised Hyperspectral Unmixing via Nonlinear Autoencoders

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***Abstract*— This work derives a novel unsupervised neural network-based scheme for unmixing hyperspectral pixels. A novel autoencoder structure was combined with a kernelization layer, mapping the mixed pixels in a higher dimensional space for easier separability, along with a novel cross-product layer to account for nonlinear mixing mechanisms. K-means clustering is utilized to estimate endmembers, and radial basis functions (RBFs) are employed to measure distances in a kernelized space to estimate abundances that provide a preliminary pixel unmixing stage to be enhanced by an autoencoder structure. A novel layer is introduced, which accounts for nonlinear mixing terms by forming pertinent cross products across the abundances of each mixed pixel. This enables accurate reconstruction of the mixed pixel adhering to different nonlinear mixing models while using the abundances and endmembers' estimates obtained via the decoding stage weights. The novel network structure is flexible since not only can it accommodate higher degree cross products of abundances and the corresponding endmember weights but it can also consider other mixing models, such as the postpolynomial nonlinear mixing (PPNM) model. Extensive testing across semisynthetic and real-world datasets shows that the proposed method, while being highly versatile in structure, also outperforms recent state-of-the-art unmixing methods.**

***Index Terms*— Autoencoders, hyperspectral imaging, neural networks, nonlinear unmixing, radial basis functions (RBFs).**

I. Introduction

HYPERSPECTRAL images, unlike traditional images that display information only across the visible light spectrum, provide information in several spectral bands [1]. These spectral bands can include the short-wave infrared (SWIR), long-wave infrared (LWIR), and ultraviolet (UV), depending on the application. Thus, such images, in which the spectra are split into hundreds of bands, display much finer information pertaining to light received from materials encompassed by these images compared to traditional images. The higher level of information can then be used in hyperspectral classification, clustering, geological exploration, crop health monitoring, and so on, with much greater accuracy [2].

The greater information across spectral bands, however, sacrifices spatial resolution, which is significantly reduced. For hyperspectral images that are taken at a considerable distance, such as aerial images of agricultural landscapes, this implies that, as the spatial resolution becomes lower for the hyperspectral image encompassing the same ground area, each image pixel encompasses a larger area. As a result, each pixel can display objects situated in that larger area [3]. Due to this caveat, it would be necessary to take a pixel containing a mixture of materials and split the mixture into light information from multiple materials.

Hyperspectral unmixing schemes usually rely on the assumption that mixed pixels come from a linear mixture of the reflectances of pure materials, also known as *endmembers*. This is true in cases where the hyperspectral images concern solid materials, as is the case with the images in this article. A large body of work revolves around the simplified yet tractable linear mixing model (LMM). A very popular method for unmixing using the LMM in an unsupervised setting is the vertex component analysis, or VCA method [3]. In this method, the pixel vectors are projected onto the feature space, where the pixels are represented by points that reside on a subspace. The endmember points are considered to be the vertices of a simplex, while the remaining points (mixed pixels) are obtained as linear combinations of those vertices. The canonical correlations' analysis has been used in conjunction with the kernel trick [6] to observe the correlations present in the data pixels mapped to a higher dimensional space in order to ascertain the abundances and use that to obtain endmember values [7]. Minimizing the volume spanned by these vertices is also studied, in conjunction with a nonnegative matrix factorization in [5]. Spatial information was also used in [4]. The work in [8] is limited in the sense that it utilizes exclusively fully connected layers, whereas more modern schemes apply a more sophisticated combination of layers (such as our novel method) and implement constraints to increase effectiveness. Feng *et al.* [9] use a constrained nonnegative matrix factorization with a sparsity constraint on the abundance matrix, while Ozkan *et al.* [10] also enforce sparsity in the abundance matrix as a means for regularization but, in order to perform effectively, rely on the assumption that the reference abundance matrix contains some sparsity as well. Different from [8] and [9], our novel framework is unsupervised and does not rely on reference data.

It has since been established that mixed hyperspectral pixels can contain nonlinear combinations of the endmembers [11]. Multiple models exist, which aim to formulate the nonlinear components in the mixtures [12]. These models have since been implemented in methods working toward hyperspectral unmixing. The work in [13] builds upon the method in [7] in

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order to optimize for a constrained cost function that relies on a nonlinear mixing model.

Neural networks have been used in recent years to carry out the task of hyperspectral unmixing. The work in [14] uses autoencoders to unmix hyperspectral images in an unsupervised manner. The work in [15] uses multiple autoencoders stacked alongside each other. This allows the final layers to extract more meaningful information from the input pixels. The ability for generative neural networks to acquire features of the mixed pixels and generate synthetic samples in order to boost the pool of available data is effectively exploited in [16]. Convolutional neural networks have also been utilized for unmixing [17]. The ability to share input data to multiple layers to boost the extraction of meaningful features, in order to compress the data effectively in smaller vectors, is considered in [18]. Nonlinear unmixing is a more recent development, with a supervised neural network setting in [19] aiming to use ground-truth information of synthetically generated mixed pixels to train the network.

Radial basis function (RBF) neural networks have also been implemented for hyperspectral classification in recent years [20], [21]. RBFs rely on effectively calculating the parameter multiplied with the Euclidean distance of sample pairs within the exponential term in order to control effectively the width of the kernel. In [20], this parameter's value is calculated via an optimization problem, where within-class entries are driven toward 1, and between-class entries are driven toward 0. In this way, the resultant estimated abundance matrix would be driven toward a sparse structure that is more suited for clustering rather than unmixing. Also, the RBF distances are measured across the input pixels with respect to each other, unlike our method where they are measured with respect to RBF centers. In [21], the RBF centers are obtained by averaging the pixel vectors belonging to the same class, which is possible for [21] given that it is supervised. For unmixing purposes, Altmann *et al.* [22] use an orthogonal least squares with an RBF layer to unmix pixels using the LMM, which, unlike our framework, is done via a supervised approach. The work in [23] is a recent work that is unique in which it attempts to use autoencoders to nonlinearly unmix pixels in an unsupervised manner. However, this scheme uses a postnonlinear mixing model [27], which is only one of the various mixing models that exist, thus limiting its versatility. Furthermore, it forces the nonnegativity constraints of the abundances by taking their absolute values. This approach is nonideal since it can lead to inaccurate abundance estimates, which will further corrupt the estimation of the endmember reflectance values.

The autoencoder networks designed in our work are versatile in their ability to unmix according to a variety of popular nonlinear unmixing models. Furthermore, they have the capability of accommodating for higher degrees of nonlinear terms that seek to account for the higher order interactions among light emanating from the hyperspectral image materials. The kernel trick is utilized to measure the contributions of the materials toward the mixed pixels in the higher dimensional space. Our novel unmixing framework also exhibits increased robustness to severe noise that may be present in the hyperspectral pixels.

This article is structured as follows. The mixing models used and the theory behind the RBF kernel applied for abundance estimation are provided in Section II. The autoencoder network designs are shown in Section III. For the three mixing models utilized in our work, the autoencoder networks concerning the Fan and bilinear models [24] are described in Section III-A1, while the network built for the postpolynomial nonlinear mixing (PPNM) model is discussed in Section III-A2. The RBF layer that was built for our work is described in detail in Section III-B. The accommodation for higher degree cross products is detailed in Section III-D. Finally, extensive numerical tests are shown in Section IV, demonstrating the robustness of the novel framework when the data are subject to severe noise and high-order nonlinearities.

II. Problem Statement and Preliminaries

Hyperspectral images contain information in hundreds of bands, thus providing a finer view of different materials being observed. Let B be the number of spectral bands across which information is obtained per pixel, while the set $\mathcal{B} := \{1, \ldots, B\}$ denotes the indices of the spectral bands. Let $\mathbf{M} \in \mathbb{R}^{N \times B}$ contain the reflectances of N pixels. One major issue in hyperspectral images is that of mixing where pixels are formed as (non)linear combinations of reflectances of pure materials: the *endmembers*.

A popular mixing model where the reflectance values of mixed pixels $\mathbf{M}$ are expressed as a linear combination of P endmembers is the LMM, i.e.,

$$\mathbf{M} = \mathbf{A} \cdot \mathbf{E} + \boldsymbol{\epsilon} \tag{1}$$

where $\mathbf{A} \in \mathbf{R}^{N \times P}$ represents the matrix of fractional contributions of the endmembers, also known as *abundances*. Matrix $\mathbf{E} \in \mathbf{P}^{P \times B}$ contains the endmember reflectances, and $\boldsymbol{\epsilon}$ represents white additive noise. From (1), the mixed pixels can be expressed individually as

$$\mathbf{M}_{i,:} = \sum_{j=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{E}_{j,:} + \boldsymbol{\epsilon} \quad \forall i \in 1, 2, \ldots, N. \tag{2}$$

Since the abundances represent the fractional contributions of endmembers, matrix $\mathbf{A}$ has nonnegative entries. Furthermore, the sum of the entries in each row of $\mathbf{A}$ will sum up to one. Similarly, the entries of matrix $\mathbf{E}$ are nonnegative being the reflectance values of endmembers across the sampled spectral bands. Thus, $\mathbf{A}_{i,j} \geq 0, \mathbf{E}_{j,:} \geq 0 \forall i \in 1, 2, \ldots, N$ and $\forall j \in 1, 2, \ldots, P$, and also $\sum_{j=1}^{P} \mathbf{A}_{i,j} = 1 \forall i \in 1, 2, \ldots, N$.

There are three main nonlinear mixing models that are often used: the Fan model, the bilinear model [24], and the PPNM model [27]

$$\mathbf{M}_{i,:} = \sum_{j=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{E}_{j,:} + \gamma_i \sum_{j=1}^{P-1} \sum_{l=j+1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{A}_{i,l} \cdot \mathbf{E}_{j,:} \odot \mathbf{E}_{l,:} + \boldsymbol{\epsilon} \quad \forall i \in 1, 2, \ldots, N. \tag{3}$$

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First, the Fan model in (3) takes cross products across different endmembers to model the nonlinear mixing part with $\odot$ representing the Hadamard product or elementwise multiplication of two arrays. The values γ_i are scaling factors that control the strength of the nonlinear components with respect to the overall mixture. Typically, these hyperparameters are defined by the user. However, our framework will treat them as unknown. Second, the bilinear model is given as

$$\mathbf{M}_{i,:} = \sum_{j=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{E}_{j,:} + \gamma_i \sum_{j=1}^{P} \sum_{l=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{A}_{i,l} \cdot \mathbf{E}_{j,:} \odot \mathbf{E}_{l,:} + \boldsymbol{\epsilon} \quad \forall i \in 1, 2, \ldots, N. \tag{4}$$

Note the similarity to the Fan model, with the one difference being that it will also include self-interacting terms in the nonlinear components, as that has been shown to be a factor to also be considered [25], [26].

Third, the PPNM [27] model will simply take the LMM mixture and sum it with the square of itself

$$\mathbf{M}_{i,:} = \sum_{j=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{E}_{j,:} + \mathbf{b}_i \left(\sum_{j=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{E}_{j,:} \right) \odot \left(\sum_{j=1}^{P} \mathbf{A}_{i,j} \cdot \mathbf{E}_{j,:} \right) + \boldsymbol{\epsilon} \quad \forall i \in 1, 2, \ldots, N. \tag{5}$$

Here, the unknown vector $\mathbf{b}$ with length N has values that determine the strength of the nonlinear components of the mixtures; thus, it is similar to the vector γ_i in (3) and (4).

Hyperspectral images are found to be *unmixed* with much greater ease if the unmixing process takes place in properly selected kernel spaces of higher dimensionality than the original data space (see details in [2], [13], and [20]). Kernels often used in practice include the linear kernel, the polynomial kernel, the Gaussian kernel [6], and so on. The Gaussian (RBF) kernel has been shown to be effective with hyperspectral images and will be employed here. In detail, the mixed pixels are represented as points in the Gaussian kernel space, and the endmembers will be treated as cluster centers in that kernel space. Thus, for a certain mixed pixel, the lower its distance is from a particular center, the closer the mixed pixel is in resemblance to that corresponding endmember, and thus, the higher the abundance will be for that endmember. The RBF distance of the mixed pixel $\mathbf{x}$ from endmember center $\mathbf{c}_i$ is written as

$$[\hat{\mathbf{a}}_x]_i = \exp(-\beta \cdot \|\mathbf{x} - \mathbf{c}_i\|^2) \quad \forall i \in 1, 2, \ldots, P \tag{6}$$

where β is a parameter that controls the spread of the Gaussian center. Since our framework is unsupervised, there will be no prior knowledge of the endmembers, and so they will be estimated using K-means [28] on the mixed pixels; thus, the Gaussian centers are initially estimated to be the cluster centers. In order to effectively visualize the concept, an example is shown in Fig. 1 where a mixed pixel is depicted in the kernel space, with three endmembers as the centers of three circular areas. As the mixed pixel is closer to endmember 1, it will have a higher abundance corresponding to that endmember. Similarly, for that mixed pixel, the abundance of endmember 2 will be lower, and the abundance for endmember 3 will be even lower. It should be noted that the areas in Fig. 1 are circular in order to depict neighborhoods using the Euclidean distance. This better illustrates the concept of the nearest neighbors' concept being implemented in the kernel space.

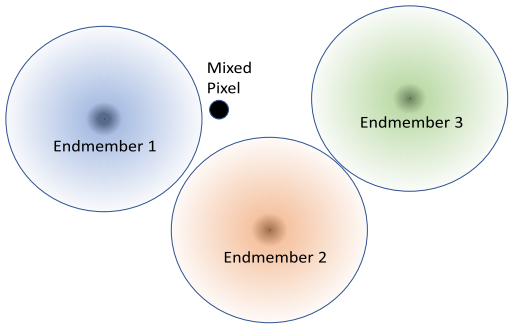


Fig. 1. Kernelized mixed pixel and endmembers' geometry.

Note that, since these distances are a measure of the abundances, the notation $[\hat{\mathbf{a}}_x]_i$ has been used, with $[\hat{\mathbf{a}}_x]$ denoting a row of the estimated abundance matrix $\hat{\mathbf{A}}$. After obtaining the P distances, normalizing them to unit summation will yield the abundance vector for this mixed pixel. It should also be noted that, since these values are a measure of distance, they cannot be negative, and thus, the nonnegativity constraint of the abundances will be automatically met.

The reason behind using K-means in conjunction with the RBF kernel trick is the similarity in their concepts. As the K-means is being performed in the feature space, it can reasonably be assumed that mixed pixels with a higher abundance toward the same endmember will encompass the same area in the feature space. Thus, the center of that area would be where the corresponding endmember will likely reside, and so K-means would be likely able to yield a point in the feature space that would be a close approximation of that endmember. This idea is similar to the area in the kernel space around which data points would reside, where they have a higher abundance of one endmember. The center of that area, which would be circularly defined by the Gaussian kernel being used, would, thus, likely be an accurate representation of the corresponding endmember. This similarity between Gaussian centers in the RBF space and the K-means cluster centers in the feature space makes the K-means a logical choice to implement in obtaining an estimation of the endmembers.

III. Unsupervised Hyperspectral Unmixing Autoencoders

A. Autoencoders for Nonlinear Unmixing

1) Fan & Bilinear Models: A novel autoencoder neural network structure is outlined next for the task of nonlinear unmixing. Fig. 2 depicts the network structure for the three different models, where it is assumed (for simplicity and without loss of generality) to have four endmembers. Each mixed pixel vector of the hyperspectral image is fed at the input layer, and the RBF distance from the four K-means estimated endmember centers is calculated. After normalizing

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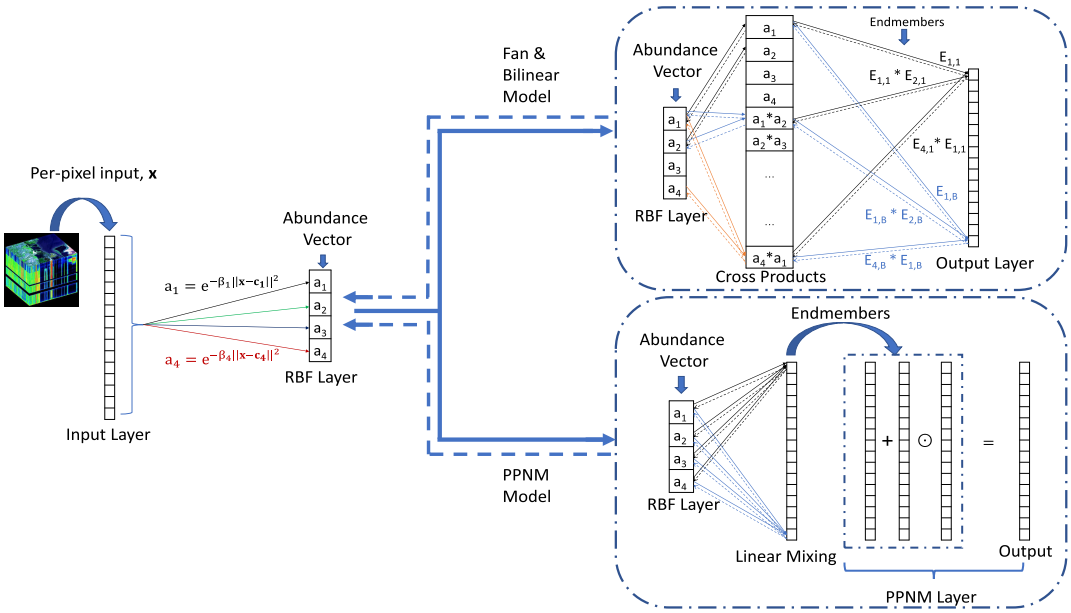


Fig. 2. Autoencoder setup for the Fan, bilinear, and PPNM mixing models (solid and dashed lines represent forward and backward propagations, respectively).

the RBF distances to unit summation, they are treated as the abundance estimates forming the RBF layer. To account for the bilinear and Fan nonlinear mixing models, a generator is introduced where cross-product terms involving the abundances are formed. The vector of abundances, along with their cross products, is then fed through the final layer, where the weights of the layer will be multiplied with the expanded abundance vector, which includes the higher order cross products to reconstruct the input mixed pixel. As the calculation of the product closely resembles the mixing models in (3) and (4), it can be observed that these weights can be interpreted as the endmembers' reflectance values and their cross products, and thus, these weights can be used to estimate the endmembers.

Although Fig. 2 (top right) depicts the cross-product terms corresponding to the Fan model, modifications to accommodate the bilinear model are straightforward by incorporating the extra self-interacting terms present in the bilinear case and adjusting the final layer weights to account for the extra terms in the cross-product generator. It should also be noted that, since the RBF centers and the weights in the final layer are both learnable parameters initialized with the K-means centers, both can actually act as estimates of the endmembers.

The autoencoder weights $\mathbf{E}$ in the final layer are trained to minimize the mean square error of some input mixed pixel $\mathbf{x}$

$$\min_{\mathbf{E},\gamma_x} B^{-1}\|\mathbf{x}-\hat{\mathbf{x}}(\mathbf{E},\gamma_x)\|_2^2 \tag{7}$$

where $\hat{\mathbf{x}}(\mathbf{E},\gamma_x)$ represents the estimated reconstruction of the input pixel at the final output layer, and γ_x refers to the value of γ pertaining to pixel $\mathbf{x}$. Thus, after obtaining the estimated abundance vector $[\hat{\mathbf{a}}_x]$ from (6), depending on whether the Fan or bilinear model is used, $\hat{\mathbf{x}}(\mathbf{E},\gamma_x)$ can take two forms. If the Fan model is used then

$$\hat{\mathbf{x}}(\mathbf{E},\gamma) = \sum_{j=1}^{P}[\hat{\mathbf{a}}_x]_j\cdot\mathbf{E}_{j,:} + \gamma_x\sum_{j=1}^{P-1}\sum_{l=j+1}^{P}[\hat{\mathbf{a}}_x]_j\cdot[\hat{\mathbf{a}}_x]_l\cdot\mathbf{E}_{j,:}\odot\mathbf{E}_{l,:}$$

whereas, if the bilinear model is used, then

$$\hat{\mathbf{x}}(\mathbf{E},\gamma) = \sum_{j=1}^{P}[\hat{\mathbf{a}}_x]_j\cdot\mathbf{E}_{j,:} + \gamma_x\sum_{j=1}^{P}\sum_{l=1}^{P}[\hat{\mathbf{a}}_x]_j\cdot[\hat{\mathbf{a}}_x]_l\cdot\mathbf{E}_{j,:}\odot\mathbf{E}_{l,:}.$$

2) PPNM Model: The network architecture when the PPNM mixing model is utilized is depicted in Fig. 2 (bottom right). The PPNM layer has the nonlinear component formed as the Hadamard product of the linearly mixed pixel summation with itself, which is added to the linearly mixed pixel.

The learnable parameters here would be nearly the same for the RBF layer. The exception is that, for the PPNM layer, the $\mathbf{b}$ values are the learnable parameters. The cost function to minimize here would be very similar to (7), with the difference of having the nonlinear coefficient γ_x being replaced by b_x. The training cost takes the form

$$\min_{\mathbf{E},b_x} B^{-1}\|\mathbf{x}-\hat{\mathbf{x}}(\mathbf{E},b_x)\|_2^2. \tag{8}$$

Similarly, the reconstruction $\hat{\mathbf{x}}$ can be expressed as

$$\hat{\mathbf{x}}(\mathbf{E},\gamma) = \sum_{j=1}^{P}[\hat{\mathbf{a}}_x]_j\mathbf{E}_{j,:} + b_x\left(\sum_{j=1}^{P}[\hat{\mathbf{a}}_x]_j\mathbf{E}_{j,:}\right)\odot\left(\sum_{j=1}^{P}[\hat{\mathbf{a}}_x]_j\mathbf{E}_{j,:}\right).$$

B. RBF Layer

For each hyperspectral pixel feeding the autoencoder neural network in Fig. 2 as input, the RBF distances from the kernelized endmember centers are calculated. As explained in Section II, the RBF distances quantify the contribution of each endmember in the input mixed pixel, thus providing an estimate for the unknown abundances. After the P RBF

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distances are obtained, they are simply normalized to unit summation, and the estimated abundances are obtained.

In the RBF layer, the learnable parameters would be the RBF centers, as well as the β value in (6). The β value basically controls the radius of the multidimensional Gaussian bell footprint area (which depicts the area in which approximately ~99% of the probability mass is present). As this radius is dependent upon the distance of the mixed pixels from the centers that they belong to, β is taken as the inverse of the squared average distance of the mixed pixels from the cluster centers that they belong to, with the centers found via K-means. This will ensure that approximately 99% of the time the mixed pixels will belong to the right center area when β is selected as described. Then, for a mixed pixel $\mathbf{M}_{i,:}$, let the label for one of the contributing classes be $L_{M_{i,:}}$ and the corresponding cluster center be $\mathbf{c}_{L_{M_{i,:}}}$. Thus, β is calculated as

$$\beta = \left(N^{-1} \cdot \sum_{1=1}^{N} \|\mathbf{M}_{i,:} - \mathbf{c}_{L_{M_{i,:}}}\|_2 \right)^{-2}. \tag{9}$$

Since the weights $\mathbf{E}$ in the autoencoder configuration represent the endmember entries, they are initialized with the K-means cluster centers of all the mixed pixels in the dataset. Even though VCA [3] is more popular as an initialization option and can be argued as a better alternative, there is one issue. With a dataset having high SNR, VCA performs endmember estimation with slightly higher accuracy; however, with low SNR, K-means is found to be more accurate. This is because VCA will choose P mixed pixels and consider them to be the best estimations of the endmembers. Not only would this be less accurate with a dataset having pixels where pure pixels do not exist but also, when dealing with data having high noise, this will compromise the endmember estimation. In addition, if severe noise distorts the endmember, there is a possibility of VCA finding multiple endmembers from one material while disregarding other materials. However, with K-means, since the centers are pixel averages within some cluster, the noise will be reduced leading to more accurate endmember estimation.

Recall that the RBF distances are nonnegative; therefore, the abundances estimated will satisfy the nonnegativity constraints. The RBF layer's output values would need to be normalized to conform to the unit summation constraint, i.e.,

$$[\hat{\mathbf{a}}_x]_i = \frac{[\hat{\mathbf{a}}_x]_i}{\sum_{j=1}^{P} [\hat{\mathbf{a}}_x]_j} \quad \forall i \in 1, 2, \ldots, P. \tag{10}$$

Then, the vector of abundances needs to be mapped to a higher dimensional space to include the cross-product vector, the length of which is dependent upon whether the Fan or bilinear model is chosen. The expansion is a straightforward method, which can be achieved without defining a new layer. This will result in the vector labeled "cross products" in Fig. 2.

C. Backpropagation

Simplifying (7), the error can be rewritten as $e := \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$. Thus, in the RBF layer, after taking the K-means cluster centers as an initial estimate of the endmembers, let the first cluster center vector be denoted $\mathbf{c}_1$. The dimension of β is $P \times 1$, where P represents the number of endmembers. Although these values are all initialized by a single scalar following the value obtained by (9), as they are weights, they can update to different values. The cluster centers have dimensions $P \times B$, where B represents the number of spectral bands. Let f_1 denote the activation function giving the first estimated abundance in the RBF layer, i.e.,

$$f_1 = \exp\big(-\beta_1 \cdot \|\mathbf{x} - \mathbf{c}_1\|_2^2 \big). \tag{11}$$

Here, β_1 will be the value of β corresponding to the first cluster center (or estimated endmember) $\mathbf{c}_1$. Performing backpropagation, the derivative of the function f_1 with respect to $\mathbf{c}_1$ is

$$\frac{\delta f_1}{\delta \mathbf{c}_1} = -2\beta_1 \exp\big(-\beta_1 \cdot \|\mathbf{x} - \mathbf{c}_1\|_2^2 \big)(\mathbf{x} - \mathbf{c}_1).$$

The derivative of the function f_1 with respect to β_1 is

$$\frac{\delta f_1}{\delta \beta_1} = -\|\mathbf{x} - \mathbf{c}_1\|_2^2 \exp\big(-\beta_1 \cdot \|\mathbf{x} - \mathbf{c}_1\|_2^2 \big).$$

For properly selected step size α, the weights during iteration *i+1* are updated using gradient descent as

$$\mathbf{c}_1^{i+1} = \mathbf{c}_1^i - \alpha \frac{\delta e}{\delta \mathbf{c}_1^i}, \quad \beta_1^{i+1} = \beta_1^i - \alpha \frac{\delta e}{\delta \beta_1^i}. \tag{12}$$

In the final layer, let the first value in the estimated pixel vector be $\hat{\mathbf{x}}_1$. Thus, following the Fan model from (3):

$$\hat{\mathbf{x}}_1 = \sum_{i=1}^{P} [\hat{\mathbf{a}}_x]_i \cdot \mathbf{E}_{i,1} + \gamma \sum_{i=1}^{P-1} \sum_{j=i+1}^{P} [\hat{\mathbf{a}}_x]_i \cdot [\hat{\mathbf{a}}_x]_j \cdot \mathbf{E}_{i,1} \cdot \mathbf{E}_{j,1}.$$

Note that γ has no subscript here as it is assumed to be a scalar for one mixed pixel. The derivatives of output $\hat{\mathbf{x}}_1$ with respect to $\mathbf{E}_{1,1}$ and γ would be

$$\frac{\delta \hat{\mathbf{x}}_1}{\delta \mathbf{E}_{1,1}} = [\hat{\mathbf{a}}_x]_1 + \gamma \sum_{j=2}^{P} [\hat{\mathbf{a}}_x]_j \cdot \mathbf{E}_{j,1},$$
$$\frac{\delta \hat{\mathbf{x}}_1}{\delta \gamma} = \sum_{i=1}^{P-1} \sum_{j=i+1}^{P} [\hat{\mathbf{a}}_x]_i \cdot [\hat{\mathbf{a}}_x]_j \cdot \mathbf{E}_{i,1} \cdot \mathbf{E}_{j,1}. \tag{13}$$

Similar to (12), the weights during iteration $i + 1$ are updated using gradient descent as

$$\mathbf{E}_{1,1}^{i+1} = \max\left(\mathbf{E}_{1,1}^i - \alpha \frac{\delta e}{\delta \mathbf{E}_{1,1}^i}, 0 \right), \quad \gamma^{i+1} = \gamma^i - \alpha \frac{\delta e}{\delta \gamma^i} \tag{14}$$

where the projected gradient step imposes the nonnegativity constraints for the weights $\mathbf{E}_{i,j}$. A similar procedure can be followed for the rest of the weights and other nonlinear models considered. Notice also from (13) that each weight $\mathbf{E}_{i,j}$ will be updated while fixing the other ones to their most recent update. Since the Adam optimizer will be used here, there is no need to explicitly define such equations for backpropagation during implementation; rather, the optimizer will account for this automatically. Nevertheless, these equations have been provided for further clarity.

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D. Higher Degree Cross Products

Earlier, a cross-product generator was introduced to calculate the nonlinear components of the mixture via second-degree products of the abundances and endmembers, thus emulating the interactions between light from the materials as they propagate toward the sensor. This is the reason why the mixing models in (3)–(5) addressed the nonlinear components via second-degree cross products only. It is possible that light from more than two materials can interact with each other, which justifies the additional calculation of higher degree products of corresponding endmembers and abundances.

In certain cases, higher degree terms can be significant enough to warrant their calculations [11]. To that end, our novel autoencoder setup in Fig. 2 for the Fan and bilinear models facilitates the incorporation of higher degree cross products. This can be done readily by stacking higher order cross-product terms right below the second-order terms while introducing additional endmember weights in the output layer. For example, when considering a nonlinear mixing model accounting for fourth-order cross-product terms, the reconstructed mixed pixel at the output of the autoencoder structure can be expressed as

$$\begin{aligned}\hat{x}(E,\gamma) &= \sum_{j=1}^{P}[\hat{\mathbf{a}}_x]_j \cdot \mathbf{E}_{j,:} + \gamma_x \\ &\times \left(\sum_{j=1}^{P-1}\sum_{l=j+1}^{P} [\hat{\mathbf{a}}_x]_j \cdot [\hat{\mathbf{a}}_x]_l \cdot \mathbf{E}_{j,:} \odot \mathbf{E}_{l,:} \right. \\ &+ \sum_{j=1}^{P-2}\sum_{l=j+1}^{P-1}\sum_{m=l+1}^{P} [\hat{\mathbf{a}}_x]_j \cdot [\hat{\mathbf{a}}_x]_l \cdot [\hat{\mathbf{a}}_x]_m \cdot \mathbf{E}_{j,:} \odot \mathbf{E}_{l,:} \odot \mathbf{E}_{m,:} \\ &+ \sum_{j=1}^{P-3}\sum_{l=j+1}^{P-2}\sum_{m=l+1}^{P-1}\sum_{n=m+1}^{P} [\hat{\mathbf{a}}_x]_j \cdot [\hat{\mathbf{a}}_x]_l \cdot [\hat{\mathbf{a}}_x]_m \cdot [\hat{\mathbf{a}}_x]_n \\ &\left. \cdot \mathbf{E}_{j,:} \odot \mathbf{E}_{l,:} \odot \mathbf{E}_{m,:} \odot \mathbf{E}_{n,:} \right). \end{aligned} \tag{15}$$

The introduction of higher degree terms can be computationally taxing. Nevertheless, the inclusion of this feature significantly adds to the proposed scheme's versatility, and results will be shown of the proposed framework's performance advantage when working with higher degree terms. Note that existing works are usually limited to bilinear terms only. Also, note that, due to the unique dynamic structure of the output layer due to this property, although the layer is similar in ways to a fully connected layer, this causes the number of nodes in the input to be variable. The PPNM model will be an exception to this fact, as that model cannot accommodate this feature. As an example, if the chosen mixing model has P endmembers and is considering up to the nth-degree nonlinear terms, then the Fan model will have $\sum_{i=2}^{n} {}^{P}\mathbf{C}_i$, and the Bilinear model will have $\sum_{i=2}^{n} i^P$ nodes in the input, where ${}^{P}\mathbf{C}_i$ denotes "P choose i."

IV. Numerical Tests

The proposed novel unmixing scheme is compared with four other methods: 1) the linear autoencoder in [14]; 2) the nonlinear autoencoder in [23]; 3) the minimum volume constrained nonnegative matrix factorization (MVCNMF) [5] method; and 4) the VCA [3] with fully constrained least squares (FCLS) for abundance calculation since VCA is used only for endmember extraction. These methods encompass all possible combinations of hyperspectral unmixing schemes that are either neural network-based or not and focused either on the linear or a nonlinear mixing model.

The work in [14] is an autoencoder network that uses mainly fully connected layers in order to take the input layer and compress it into a latent vector that represents the abundances. For the activation functions in the layers of this network, the leaky rectified linear unit (ReLU) was used. This method will be referred to as the linear neural network (LNN). The work in [23] is an autoencoder network that performs unsupervised nonlinear unmixing, and it utilizes the PPNM mixing model. Here, the ReLU activation was used for the layers. This method will be referred to as the non-LNN (NLNN). The work in [5] is an unsupervised scheme that aims to minimize the volume spanned by the endmembers and uses that as a constraint to perform nonnegative matrix factorization for unmixing.

The tests were conducted using both semisynthetic and real datasets. For the semisynthetic datasets, three source datasets were used: the Pavia Center, Botswana, and Pavia University datasets [1]. Four classes from each dataset were chosen, and for every 1000 pixels, one class had a majority abundance of between 80% and 90%, which was randomly chosen. The remaining abundances were randomly split between the other three classes. Doing this for every class, each dataset had a total of 4000 pixels. During the creation of the data, for the Fan and bilinear models, the values for γ_i were set to 1, while, for the PPNM model, the values of $\mathbf{b}_i$ were following a uniform distribution within the interval $[-0.3, 0.3]$. For the VCA + FCLS method specifically, for the Fan and bilinear models, the values for γ_i were presumed to be known, while, for the PPNM model, the values for $\mathbf{b}_i$ were assigned by a different set of randomly generated values using the same uniform distribution, within the same interval $[-0.3, 0.3]$. This implies that the VCA + FCLS method has an advantage over the other methods in the PPNM model.

As mentioned before, the K-means initialization of the endmember weights is a better choice compared to the well-used VCA when dealing with significant noise. To further challenge the methods used, the data were contaminated with white additive noise such that the signal-to-noise ratio (SNR) values for the data were ranging from 0 to 20 dB. For all datasets, all SNR values, and all mixing models, each accuracy metric was obtained by averaging across 50 independent trials.

The autoencoder used the Adam optimizer [30] implementing the updating recursions in (12) and (14) to learn weights that would yield an output that minimizes the root mean square error (RMSE) between the output and the input. The number of epochs was set to 50, and the learning rate was set to 10^{-4}. For the Fan and bilinear models, the learnable weights for γ were initialized to be equal to 1, while the learnable weights for values of the vector $\mathbf{b}$ in the PPNM model were initialized to be equal to 0. The weights for the RBF centers

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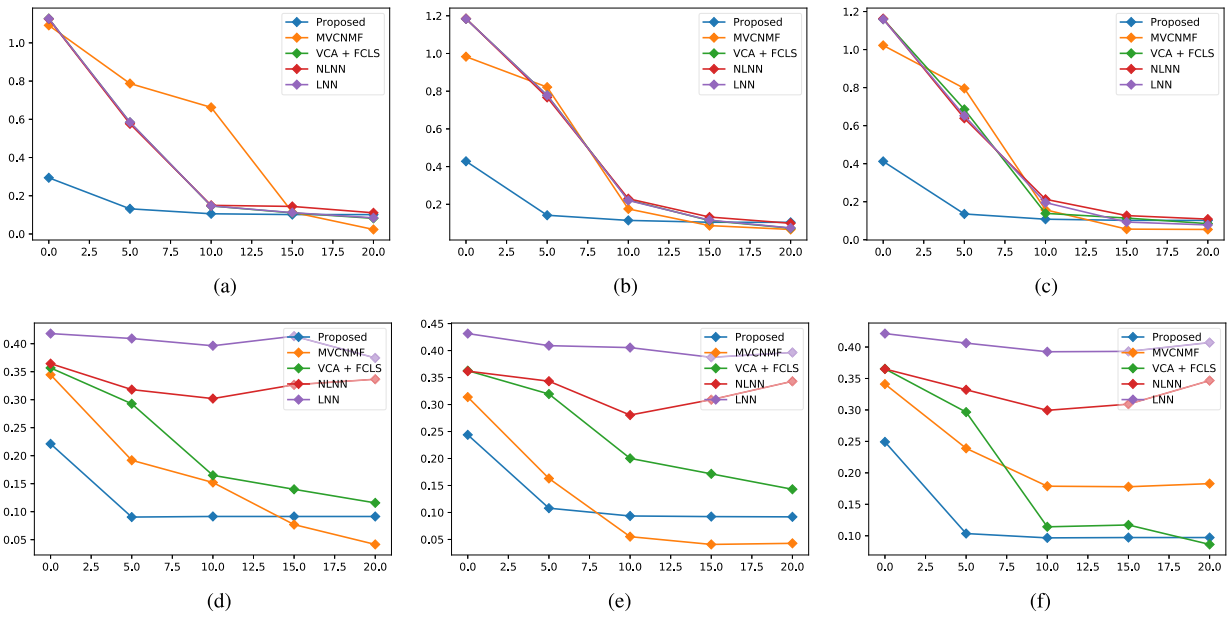


Fig. 3. SNR (dB) versus SAD and SNR (dB) versus RMSE in all methods for the Pavia University dataset. (a) SNR versus SAD in the Fan model. (b) SNR versus SAD in the bilinear model. (c) SNR versus SAD in the PPNM model. (d) SNR versus RMSE in the Fan model. (e) SNR versus RMSE in the bilinear model. (f) SNR versus RMSE in the PPNM model.

in the RBF layer, as well as the weights in the final layer for the Fan and bilinear models, were initialized to be equal to the Kmeans cluster centers of the dataset. The tests were conducted on a PC with an i5 6400 core processor with an Nvidia RTX 2060 GPU in Tensorflow 2.1.0. Since the centers in the RBF layer and the weights in the final layer can both be used as endmember estimations, either can be used to measure endmember estimation accuracy. However, the weights in the final layer form more accurate endmember estimates as they interact directly with the nonlinear terms of the abundances. The novel framework code is available at [31].

To measure endmember estimation accuracy, the spectral angle distance (SAD) metric was used. For some actual endmember $\mathbf{e}$ and its estimation $\hat{\mathbf{e}}$, the SAD is

$$\text{SAD} = P^{-1} \sum_{i=1}^{P} \cos^{-1}\left(\frac{\mathbf{e}_i{}^T \cdot \hat{\mathbf{e}}_i}{\|\mathbf{e}_i\|_2 \cdot \|\hat{\mathbf{e}}_i\|_2}\right). \tag{16}$$

For abundance estimation, the metric used was the RMSE found as

$$\text{RMSE} = (P \cdot N)^{-1} \sum_{i=1}^{N} (\|\mathbf{a}_i - \hat{\mathbf{a}}_i\|_2) \tag{17}$$

where the actual abundance for some mixed pixel is $\mathbf{a}$, and its estimation is $\hat{\mathbf{a}}$.

A. Semisynthetic Data

For the generation of semisynthetic data, the AVIRIS airborne sensor acquiring hyperspectral images over Pavia University and Pavia Center, Pavia, Italy, and a region over the country of Botswana was used [1]. The batch size was set to 4000. This is convenient in the sense that it allows for faster operations while giving a smoother reduction of the cost function over the epochs. The reason is that the gradient descent calculates the gradients for all the samples in a batch and takes the average of those gradients when updating the weights. If the batch size is small, then gradient descent moves toward a direction that optimizes taking into consideration only samples in the batch. Since samples in subsequent batches can cause the gradient to descend in a different direction, the resultant gradient descent updates have a higher likelihood to fluctuate and have an erratic pattern. Contrary, larger batch sizes will allow for a smoother and more stable descent over the entire dataset. It should be noted that, for simplicity, the results obtained here were with the assumption of prior knowledge of the correct mixing model during the unmixing process since extensive testing showed that choosing a different unmixing layer produced a negligible difference. Although this means that the network could have been designed with only one mixing model, the three major mixing models have, nevertheless, been incorporated in order to have a highly versatile network architecture.

1) Pavia University: Fig. 3 shows the plots of the SAD and RMSE values for the various methods in the Pavia University dataset. It can be seen that, for low SNR, VCA will be susceptible to the high noise, and its accuracy will be quite low compared to the proposed method. For higher SNR, given the high effectiveness of VCA, it will have a slight edge over K-means as far as endmember estimation is concerned. However, the key advantage of the proposed method lies in its ability to achieve higher separability in the kernel space, given that some hyperspectral materials might have properties that are too similar in the feature space for VCA to find correct endmembers. Due to this, VCA may erroneously ignore materials and, instead, take multiple endmembers from the same material instead. Although VCA might still be better than K-means for endmember estimation in high SNR values, VCA can perform worse during abundance estimation

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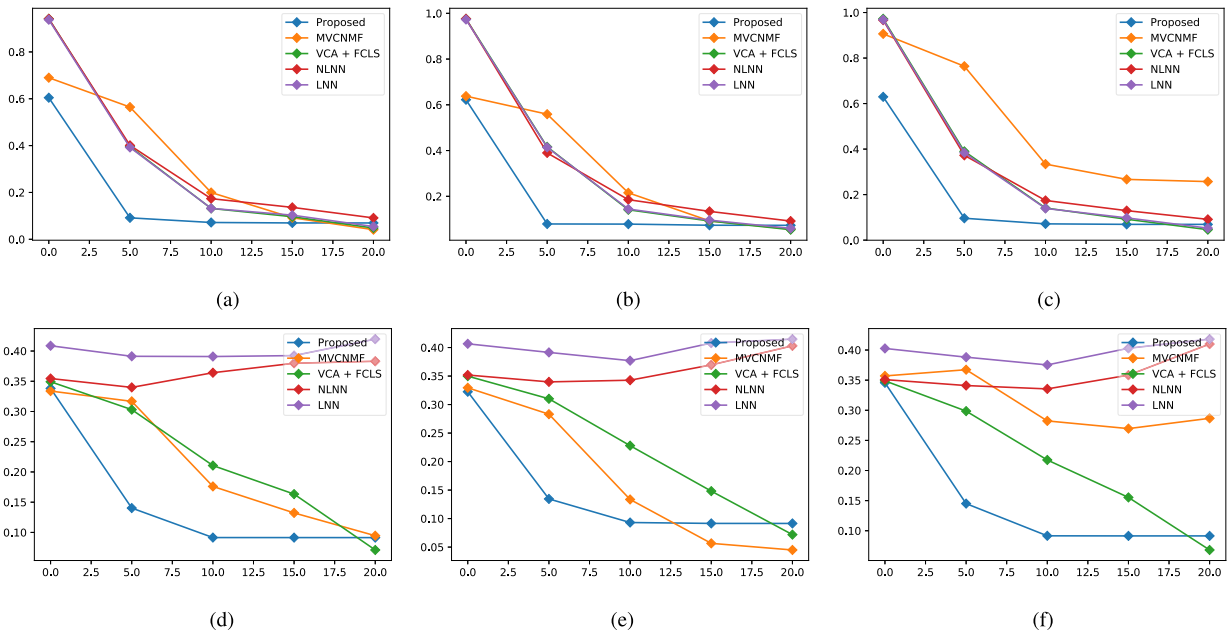


Fig. 4. SNR (dB) versus SAD and SNR (dB) versus RMSE in all methods for the Botswana dataset. (a) SNR versus SAD in the Fan model. (b) SNR versus SAD in the bilinear model. (c) SNR versus SAD in the PPNM model. (d) SNR versus RMSE in the Fan model. (e) SNR versus RMSE in the bilinear model. (f) SNR versus RMSE in the PPNM model.

since the higher abundances might be attributed to the wrong endmember. This is observed in the RMSE, as the LNN and NLNN methods face considerable difficulty. VCA + FCLS and MVCNMF have a slight edge over the proposed method in higher SNR values, but, depending on the mixing model, this edge is inconsistent.

Furthermore, it is likely that the LNN and NLNN are facing further difficulties due to the structure of using fully connected layers to obtain a latent vector that is assumed to contain the features of the endmember materials in a highly compressed fashion. Although this can be effective, it can be highly sensitive to the nature of the materials involved and will be vulnerable if the spectral reflectances of different materials are too similar. Evidently, the kernelized pixels will be facilitating better the separation of features of even challenging data, such as this, and will yield much better accuracies with greater consistency than MVCNMF and VCA + FCLS.

2) Botswana: Fig. 4 shows the accuracy measures of the various methods versus SNR values. The proposed method will either have extremely better values or noticeably better values until 15-dB SNR; then, the performance of our method will be slightly behind the MVCNMF and the VCA + FLCS methods. The RMSE is a different matter, however. Not only are the LNN and NLNN methods considerably inaccurate, but MVCNMF also faces significant difficulty for the PPNM model, while VCA + FCLS fares a little better for higher SNR. This dataset is particularly challenging for 0-dB SNR, as shown by the fact that the proposed method's SAD and RMSE are almost as high as the other methods. However, for 5-dB SNR, the proposed method is far better at accurately unmixing the pixels. In fact, above 5-dB SNR, the SAD values plateau, showing that it is performing at its peak even in severe noise. The RMSE does not trail far behind as well, reaching its peak accuracy at 5 dB.

3) Pavia Center: The third semisynthetic dataset to test is the Pavia Center. This was a particularly interesting dataset because the materials in this dataset exhibited similarities that were too high for VCA to discern well. Fig. 5 depicts that this caused high inconsistencies even in higher SNR values, especially in the 15-dB value. In fact, across the 50 iterations being run, nearly all of them caused especially high SAD and RMSE values, as can be observed in the plots. For SAD values, only MVCNMF had slightly better accuracy than the proposed method for 20 dB. However, for RMSE values, except only for the bilinear model at 20 dB compared to MVCNMF, the proposed method not only achieved the best accuracies for all values of SNR; it began to plateau at 5 dB, further showing its extremely high effectiveness in the presence of high noise. Note that the SAD of the proposed method is nearing the plateau at 0-dB SNR, showing its robustness to severe noise.

4) Higher Degree Cross Products: As mentioned previously, the proposed framework is versatile for easily allowing high-degree nonlinear terms. To this end, as there were four classes making up the various mixed pixels, the Fan model was expanded upon, and allowed for up to third degree, and also up to fourth-degree cross products [see (15)].

The plots in Fig. 6 show that there is little to no difference in the RMSE values. There is, however, a small difference in SAD values for the Pavia University and Pavia Center datasets.

Since the higher order cross products have little effect on the unmixing performance results, the experiments were rerun with a higher magnitude of γ, raising its value from 1 to 10, thus making the nonlinear component much more significant in the mixture. Despite not changing the initial value of the weights pertaining to γ in the final layer of our neural network, it was clearly the most robust with respect to the new mixtures. Fig. 7 clearly indicates that the other

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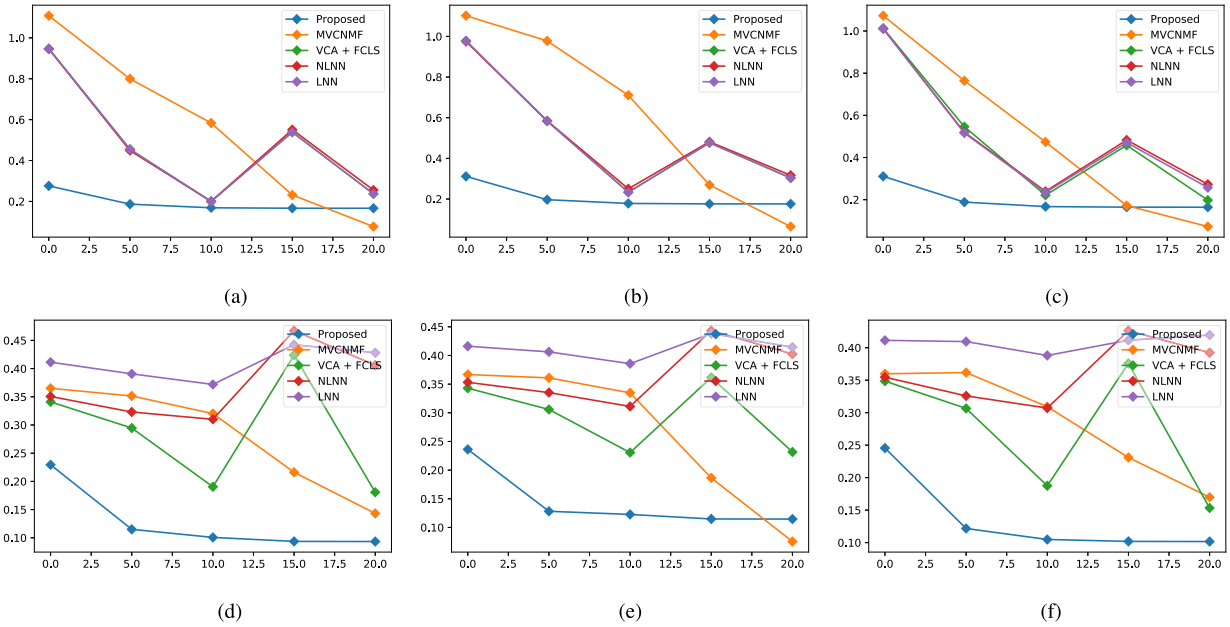


Fig. 5. SNR (dB) versus SAD and SNR (dB) versus RMSE in all methods for the Pavia Center dataset. (a) SNR versus SAD in the Fan model. (b) SNR versus SAD in the bilinear model. (c) SNR versus SAD in the PPNM model. (d) SNR versus RMSE in the Fan model. (e) SNR versus RMSE in the bilinear model. (f) SNR versus RMSE in the PPNM model.

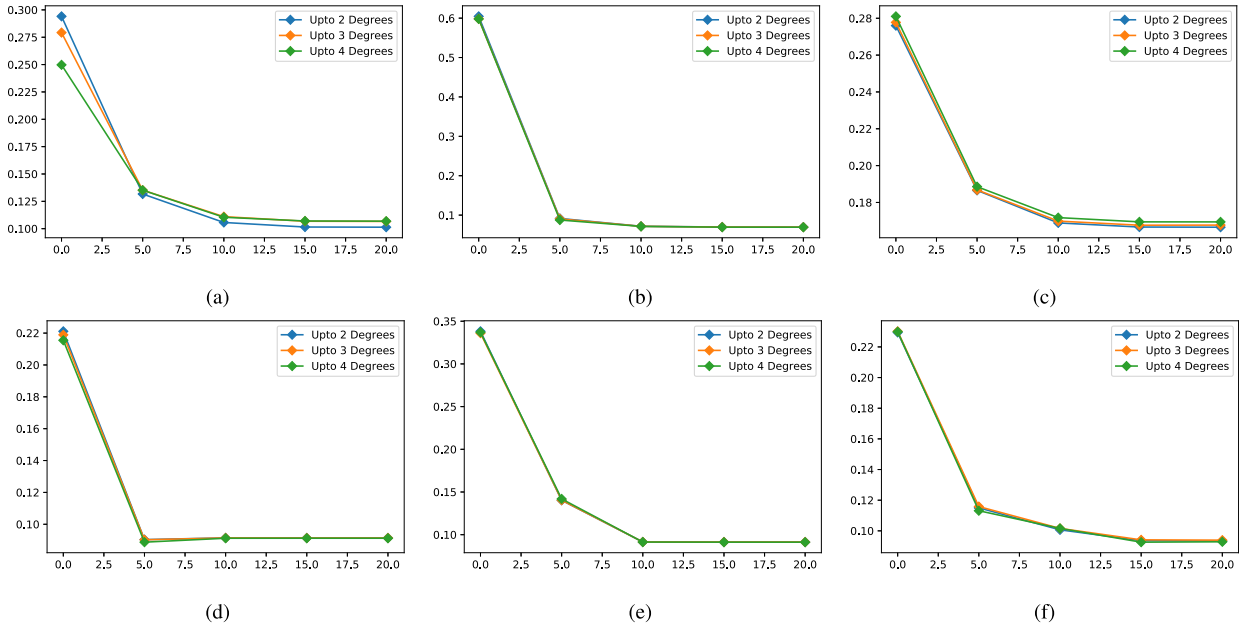


Fig. 6. SNR (dB) versus SAD and SNR (dB) versus RMSE of the proposed method with varying maximum cross-product degrees. (a) SNR versus SAD in the Pavia Univ. (b) SNR versus SAD in Botswana. (c) SNR versus SAD in the Pavia center. (d) SNR versus RMSE in Pavia Univ. (e) SNR versus RMSE in Botswana. (f) SNR versus RMSE in the Pavia center.

methods, especially the MVCNMF method, were impacted with the now-stronger nonlinear component in the mixture. However, the performance of our proposed method is largely unaffected.

5) K-Means Versus VCA Initialization: In all existing neural network-based unmixing schemes where weights are being used for endmember estimation, VCA will usually be chosen for initializing. Differently, here K-means was utilized. The primary reason is for dealing with severe data noise. As seen in the semisynthetic results, the competing methods using VCA for endmember initialization had higher SAD values than the proposed method, only in the highest SNR values, and even then those superior scores were not consistent among all the datasets. In order to further illustrate the advantage of K-means initialization in our scheme over VCA, K-means was replaced with VCA instead; then, all the previously conducted

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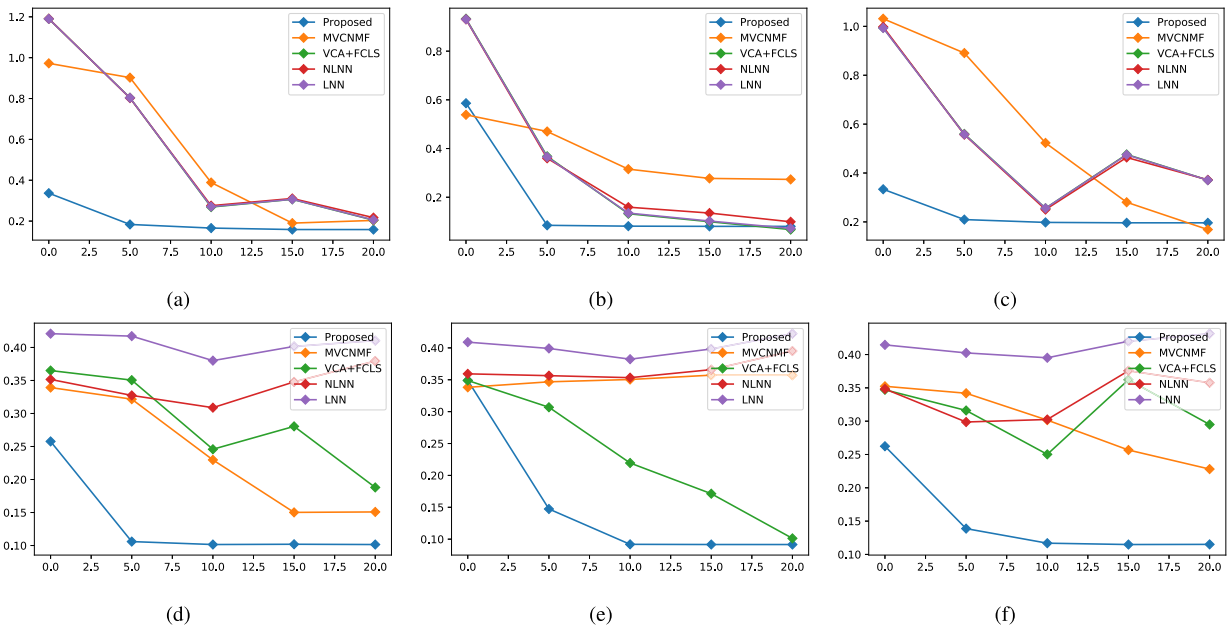


Fig. 7. SNR (dB) versus SAD and SNR (dB) versus RMSE for all methods with up to the third-degree cross products with $\gamma = 10$. (a) SNR versus SAD in Pavia Univ. (b) SNR versus SAD in Botswana. (c) SNR versus SAD in the Pavia center. (d) SNR versus RMSE in Pavia Univ. (e) SNR versus RMSE in Botswana. (f) SNR versus RMSE in the Pavia center.

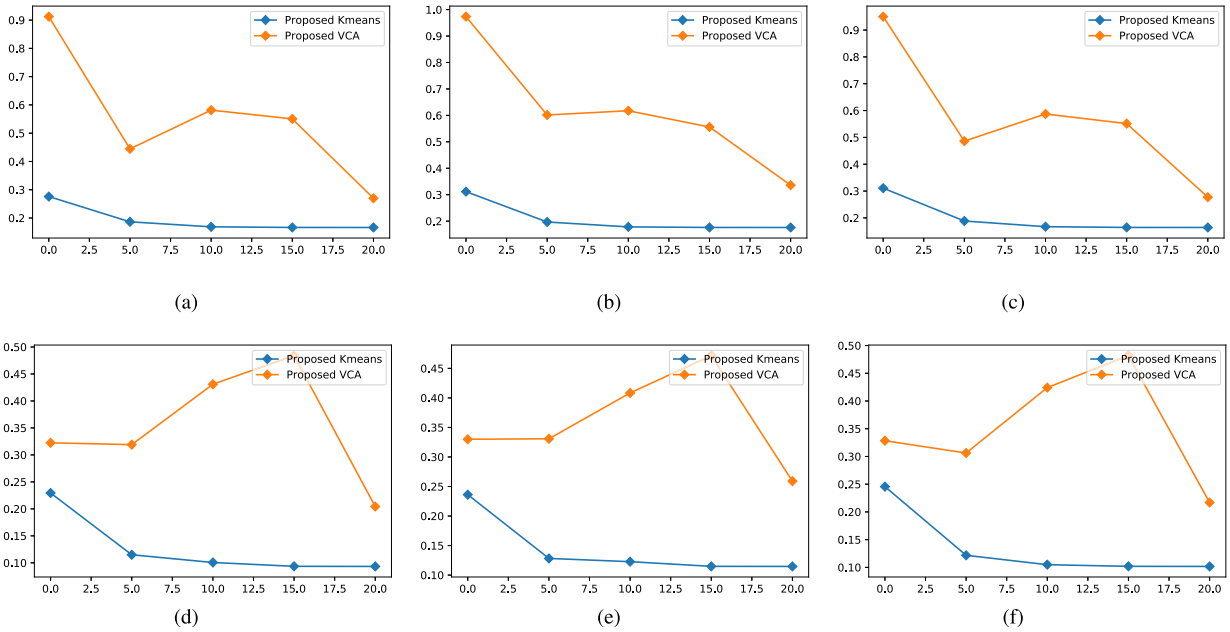


Fig. 8. SNR (dB) versus RMSE and SAD in the novel method for the Pavia Center with varying endmember weight initializations. (a) SNR versus SAD, the Pavia Center, and the Fan model. (b) SNR versus SAD, the Pavia Center, and the bilinear model. (c) SNR versus SAD, the Pavia Center, and the PPNM model. (d) SNR versus RMSE, Pavia C., and the Fan model. (e) SNR versus RMSE, Pavia C., and Bilin. model. (f) SNR versus RMSE, Pavia C., and PPNM model.

tests were rerun for 50 trials each. The plots in Fig. 8 show the advantage of having K-means as the choice of initialization over VCA for the Pavia Center dataset.

It should be mentioned that this advantage of K-means over VCA is not effective solely with the proposed method. To show how robust K-means is over VCA for severe noise, the NLNN method was rerun such that the initialization of the endmember weights was done with K-means instead of VCA for the Pavia Center dataset. Fig. 9 shows that, when initializing the NLNN method's endmember weights with K-means instead of VCA, the SAD values have an advantage over VCA in most cases for SNR values of 5 dB and every case for 0 dB. Furthermore, in the case of RMSE values, the novel framework achieves better accuracy over nearly all SNR values and over all mixing models.

6) Choice of Unmixing Layer: In the performance results provided above, the unmixing layer was always chosen to be the same as the mixing model for generating the semisynthetic datasets. However, through extensive testing, it was found that, even if the unmixing layer is different from the mixing

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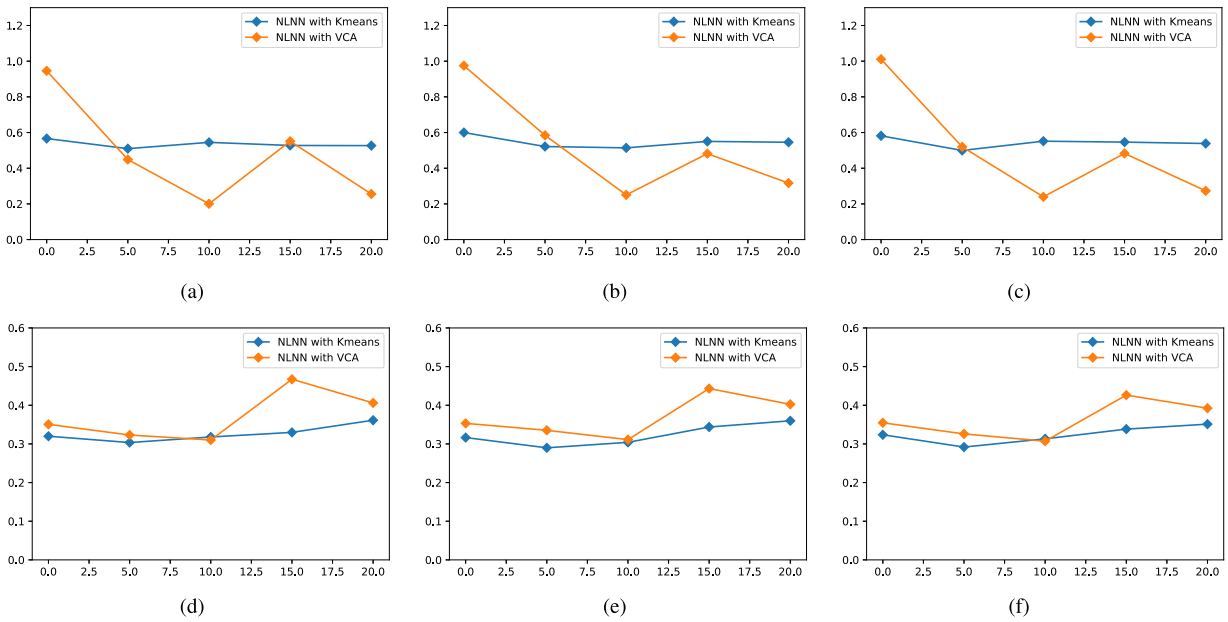


Fig. 9. SNR (dB) versus SAD and SNR (dB) versus RMSE in the NLNN [23] method for the Pavia Center with varying endmember weight initializations. (a) SNR versus SAD, Pavia C., and the Fan model. (b) SNR versus SAD, Pavia C., and the bilinear model. (c) SNR versus SAD, Pavia C., and the PPNM model. (d) SNR versus RMSE, Pavia C., and the Fan model. (e) SNR versus RMSE, Pavia C., and the bilinear model. (f) SNR versus RMSE, Pavia C., and the PPNM model.

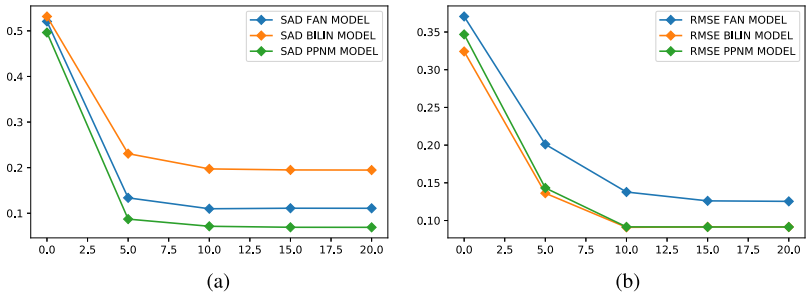


Fig. 10. (a) SNR (dB) versus SAD and (b) SNR (dB) versus RMSE in the proposed method for Botswana, with only PPNM unmixing layer against various mixing models having $\gamma = 50$.

model, the difference in performance was negligible. Indeed, this was the case when the nonlinear terms during the mixing are noticeably weak compared to the linear terms. On the other hand, the stronger the nonlinear terms are, the worse the performance will be when the unmixing model is chosen incorrectly. As an example, consider the Botswana dataset where the mixed pixels are generated with $\gamma = 50$. In Fig. 10, we can see that, if the nonlinear components are extremely strong, the correct choice of the unmixing layer becomes significant. Thus, in the case of the datasets used previously, even though the proposed method will be highly accurate regardless of the unmixing layer used, it will still be extremely effective as long as the nonlinear terms are not significantly amplified with respect to the linear terms. Under such a setting in Fig. 7 (where $\gamma = 10$), existing alternatives behave much worse than our method. Nonetheless, when nonlinear terms are strong, knowledge of the mixing model is required.

B. Real-World Data

1) Samson: For emulating real-world scenarios, the well-used Samson dataset [29] was used for testing. This is a 95×95 pixel image of a waterside area. Fig. 11 shows the heatmaps for the "*Soil*," "*Tree*," and "*Water*" classes. The NLNN method faces considerable difficulty with the "*Soil*" class of the data, and the LNN method found it challenging to tackle the same class. In the "*Tree*" class, the outer fringes of the corresponding areas provided a challenge for the proposed novel method, but it showed the best balance among all the three classes. In the lower left portion of the image, a small area proved difficult to correctly classify as part of the "*Tree*" class; instead, it was misclassified as belonging mostly to the "*Water*" class.

For VCA + FCLS, the PPNM mixing model was used, while, for the proposed method, the Fan model was employed.

2) Jasper Ridge: Another dataset to observe is the Jasper Ridge dataset [29]. This is a 100×100 pixel image of an area with a body of water through the middle. Fig. 12 shows the heatmaps for the "*Tree*," "*Water*," "*Dirt*," and "*Road*" classes. Although all the classes fared well with the "Water" class, the MVCNMF method found higher abundances for water in the rest of the image. The proposed method was significantly accurate with the "Dirt" class and was also the most accurate with the "Road" class, with MVCNMF

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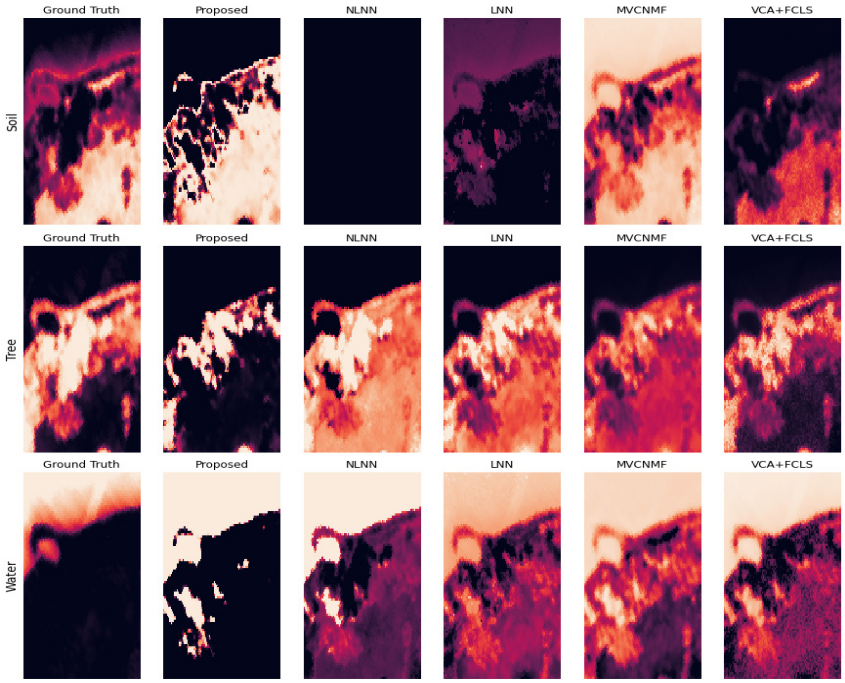


Fig. 11. Heatmaps of the Samson dataset for different unmixing methods.

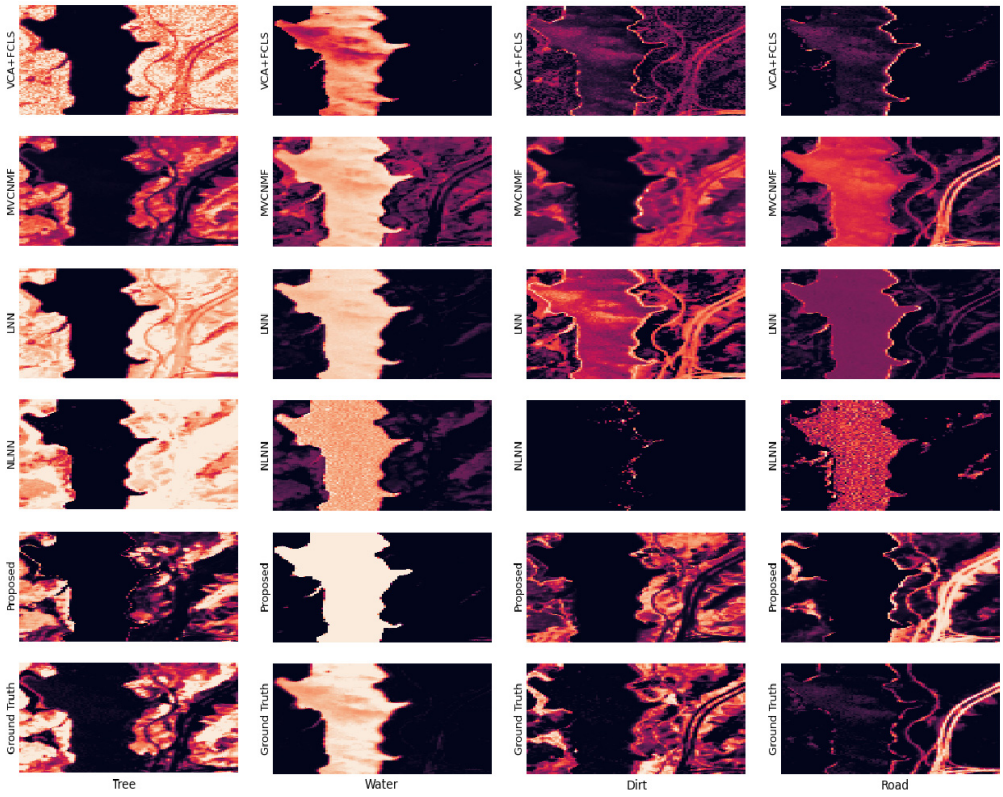


Fig. 12. Heatmaps of the Jasper Ridge dataset for different unmixing methods.

identifying the road as well, but also considered the body of water to partly contain the same material as well. In the "Tree" class, some areas had significantly high abundances in the ground truth, characterized by the highly bright areas. Although the MVCNMF did relatively well with this class, the proposed method fared the best particularly in classifying these brighter areas. Overall, the proposed method clearly performed the best.

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V. Conclusion

A novel unsupervised NN-based scheme was designed for nonlinear hyperspectral unmixing. The ability to accommodate three key mixing models, as well as the ability to consider higher degree cross products in the network, added to the scheme's high versatility. Assuming knowledge of the number of endmembers, as is typical in hyperspectral unmixing, the property of hyperspectral image materials exhibiting higher separability in the kernel space was effectively utilized toward estimation of abundances, which then led to more accurate endmember estimation. Even in extreme noise, our proposed method was significantly more effective compared to other (non)neural network-based schemes. Extensive testing with semisynthetic and real-world datasets attested to the superior performance of the proposed method over existing alternatives. Future research includes automatic unmixing model selection in the challenging case of strong nonlinear mixing components.

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